

Towards Equitable Futures in Frontline Health: Design of Intelligent Systems for Supporting (Gendered) Care Work in Resource-Constrained Settings

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ABSTRACT

Frontline health workers in many countries are responsible for filling gaps in essential primary health infrastructure, as witnessed during the COVID-19 pandemic. Their work increasingly involves the use of purportedly “intelligent” systems or data collection for such systems, to support diagnosis, disease forecasting, and information delivery. My research aims to inform the design of data-driven and automated systems in frontline health work, particularly for women workers in low-level and precarious roles in the Global South. Drawing from literature in the fields of human-computer interaction (HCI), gender and development studies, and health informatics, I will critically examine health workers’ experiences and relationships with “intelligent” systems, and engage in the participatory design of technology that might better serve worker needs while strengthening the frontline health ecology overall.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

HCI4D; AI; Future of Work; Healthcare; India

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1 CONTEXT AND MOTIVATION

The discipline of computing has been making transformative advancements in data-driven fields such as Artificial Intelligence (AI) and Machine Learning (ML), as we witness unprecedented datafication of the societies we inhabit. This is also true for the field of Global Health, where leading organizations, alongside researchers in AI and medicine, are increasingly proposing promising applications of AI in healthcare [14]. For these proposed applications to be

successful in resource-constrained contexts such as in parts of the Global South, they must align with and strengthen existing frontline health infrastructures that deliver critical last-mile care. To this end, my dissertation research aims to introduce and augment technologically mediated frontline health workflows while taking a participatory design approach, to enable variously healthier futures. My research will also critically examine the push for “intelligent” systems in frontline health, and health workers’ relationships and experiences with such systems [14], focusing our attention on computing technologies characterized by autonomy, the ability to learn, and the ability to interact with other systems and with humans.

In many regions of the Global South, frontline health workers (FHWs) are responsible for last-mile delivery of care. In India, where my dissertation research takes place, the FHWs are predominantly women with little medical training, tasked by healthcare authorities with providing reproductive, maternal, and newborn care in resource-constrained settings. These FHWs are key community actors who deliver additional kinds of care work as well: they service the information needs of women and girls across ages, provide general wellness information to households, act as information liaisons between the community and higher authorities, and more. The essential nature of their work has also been amply demonstrated through the COVID-19 pandemic, which has drawn attention to these workers’ efforts and challenges in numerous ways. Yet, FHWs remain under-recognized, overburdened, and underpaid, which directly impacts the quality of care received by communities on the margins. These aspects of their workflows have only recently begun to receive focused attention in the context of technology design.

As state and healthcare authorities increasingly digitize and automate workflows, there is a risk of technology perpetuating inequities for workers and communities. Despite claims to streamline workflows, these systems largely overlook routine work performed by FHWs, lack transparency, and offer little utility to workers and target communities [12]. I will explore an alternative set of values in data-driven healthcare provision by centering the perspective of workers. My dissertation research will investigate how data-driven systems might be designed to support frontline health workflows by selectively automating routine and burdensome workflows and enabling better distribution of work.

2 RELATED WORK

Below I outline related literature on the role of technology in frontline health, AI for frontline health, and the emerging area of HCI for AI.

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2.1 Technology in Frontline Health

The field of Information and Communication Technologies and Development (ICTD) has long been invested in the design, development, and deployment of technological interventions across the Global South. Since the early 2000s, ICT interventions have been deployed to support maternal health [21], sexual health [15], and tuberculosis [2]. Early ICTD research by Ramachandran et al. studied the use of mobile phones by FHWs for information delivery [36]. Since then, several studies have focused on information dissemination by FHWs through videos, around maternal and child health [23, 27]. Kumar et al. have also studied how the content itself could be locally produced and designed to be locally relevant [23]. In follow-up work, Vashistha et al. studied the learning gains among FHWs as a result of this intervention [44]. In related efforts, Yadav et al. studied how peer learning could be facilitated among FHWs, using IVR and mobile applications [48, 50]. There have also been efforts to directly connect with communities, with or without keeping FHWs in the loop. For instance, Okeke et al. designed a USSD-based system to elicit feedback from care recipients [29]. Perrier et al. used SMS to deliver health information to pregnant women as well as their male partners [33, 34].

Data-driven health research efforts have largely been directed towards supporting surveying and information dissemination by FHWs. Research in this area has considered the use of mobile devices for digitizing data collection and improving the efficiency of current processes [5, 24], and the affordances and constraints of paper and mobile surveys [3]. DeRenzi et al. have studied how this data can be used to improve workflows and provide FHWs with feedback on their performance [4]. Researchers have also uncovered the sources of tension when digitizing data collection by FHWs [12, 31]. This body of work has highlighted the potential for technology to add legitimacy to the work of FHWs who are underpaid and under-recognized [21, 36], the value offered by mobile phones used by FHWs beyond work [13, 24], and interest among FHWs to improve their digital literacies [13, 31]. Recent participatory design research by Yadav et al. has also begun to explore how artificial intelligence systems can play a role in such contexts, such as the use of chatbots to support breastfeeding practices [49]. My work further builds on this body of work.

2.2 The Future of Health Work

The design of technology for health work is an active area of interest for scholars in the fields of HCI and Computer-Supported Cooperative Work (CSCW). The entry of data-driven technologies in healthcare (as in other domains), further precipitated by the COVID-19 pandemic, has led to a shift in the nature of work and the workplace [32]. Work is increasingly online, automated, and paid on per task basis (also known as gig work). Researchers have documented the effects of AI systems, automation, and gig work on the workforce [6]. For instance, Gray and Suri have uncovered the invisible work that goes into making AI systems work, and the underpaid and exploitative nature of this work in the corporate sector [8]. Thakkar et al. have described how many vocational workers are unprepared for the automation of their work, and are excluded from current technological platforms for skilling and job-seeking [43]. Within healthcare, technology for health work has typically focused

on hospital settings, such as for clinical decision support (e.g. in cystic fibrosis [38] and intensive care [16]), and to enable coordination across nurses, doctors, and patients. For example, researchers have studied how technology could support collaboration between care teams and patients with cancer [42], enable communication across hospitalized patients and caregivers [26], and documented the use of chat for work by nurses in China and India [17, 45]. More recently, HCI researchers have studied informal care outside clinical settings, such as in families [35, 40] and communities [18], and have proposed the design of technology to support the invisible social and emotional work of caregiving. Researchers have also studied how artificial intelligence could support the routine, “unremarkable” work of clinicians [51], and help with collaborative decision making across medical experts [1].

In each of the above cases, some work and workers are more visible and better supported than others. In general, work that is gendered (e.g. care work [7, 52]) or is viewed as unskilled is less supported. Frontline health, in particular, makes up the lowest tier of healthcare work and is primarily comprised of women workers [19]. As feminized work conducted largely by and for women from marginalized groups, FHWs remain underpaid. Their care work, data work, and physical labor are seen as unskilled, and they have few opportunities for professional growth [12]. My proposed research will explore how technology can lead to better futures for such workers by—making work visible, and supporting the distribution and automation of invisible work.

2.3 Gender and Digital Access in the Global South

HCI and ICTD have long studied mobile adoption and use in the Global South, with a view to understand how marginalized populations are gaining access to the increasingly ubiquitous mobile phone and making meaning of it across disparate contexts. Smyth et al.’s study of entertainment-driven mobile use in urban India [41], Mehra et al.’s most recent work that updates our understanding of these mobile phone ecologies [25], Pal’s analysis of the symbolism of technology in India [30], Rangaswamy and Cutrell’s research on mobile adoption by urban Indian male youth [37], Nemer’s ethnography uncovering how marginalization is propagated online [28], and Kumar and Parikh’s work on the changing materiality of digital music [22]—these are just some of the many studies that have brought focus and attention to the multi-faceted and increasingly global phenomenon of mobile appropriation “on the margins”.

With women from marginalized backgrounds increasing their adoption of mobile technologies (and more recently, the internet), this phenomenon too has received much attention in HCI4D literature in recent years. Early work on women’s mobile practices in Ghana [47], Kumar’s research on (mis)perceptions around the gender divide in India based on a study of mobile phone practices [20], Wyche et al.’s study of what women use their mobile phones for in rural Kenya [46], Ibtasam et al.’s inquiry on the financial inclusion of women in Pakistan [10], and Sambasivan et al.’s recent work on privacy and security challenges that South Asian women face in their use of mobile devices [39]—these works collectively highlight that women in the Global South are faced with a slew of additional

barriers impacting their use of the mobile phone. They are frequently the last adopters of mobile devices within their households, they depend on the male members (or children) in their family for access to mobile media, device access has typically been shared, they are generally seen as non-users even when they are not, and there are additional privacy concerns, among other gender-specific challenges that they face. In addition to signaling the generally disempowered nature of women's technology use in the Global South, these works also highlight that the landscape is changing, and mobile adoption is growing, even amidst this otherwise marginalized group. We find this as well, with our first account of these women on the margins leveraging various mobile affordances to fulfill the numerous roles that they perform on an everyday basis.

3 RESEARCH GOALS

Research Questions: I will investigate the following research questions over two studies, through a participatory approach. In the last three years, I have fostered relationships with FHWs, communities, and non-profit organizations in underserved settings in the states of Delhi and Haryana in India, which I will leverage in my research.

- (1) RQ1: *How might routine workflows of FHWs be automated (e.g. by enabling follow up visits on chat), to make work more manageable and fulfilling?*
- (2) RQ2: *How might data collected routinely be leveraged to enable better distribution of work across state and healthcare authorities?*
- (3) RQ3: *How might emerging digital tools respect worker autonomy and integrate into the frontline health ecosystem overall?*

4 RESEARCH SUMMARY

4.1 Study 1: Automating Routine Workflows on Chat

RQ1: How might routine workflows be automated (e.g. by enabling follow up visits on chat), to make work more manageable and fulfilling?

Motivation: I will explore how FHWs' work could be distributed more effectively by automating routine workflows on chat, and engaging family members in care work. I will leverage the widespread use of WhatsApp for communication between FHWs and target families in India [13]. Focusing on antenatal care, I will ask the following questions: How are FHWs and new and expecting mothers already using chat for care provision? How might we automate routine work like follow-ups by FHWs on chat? How can chat platforms be designed to engage families in antenatal care? How does automation and family engagement impact the allocation of work time by FHWs, particularly for households that need additional attention and support?

Study Design: The study is divided into two phases that involve the following: (1) interviews and surveys with FHWs and mothers on their current use of telehealth systems (complete), and (2) deployment of a semi-automated WhatsApp chatbot with FHWs and families with new and expecting mothers (system development in progress). Phase II will engage 45 families of new and expecting

mothers supported by 15 FHWs. The family members will be engaged through a WhatsApp chatbot over four months. The focus is on family members because of the high burden of care work on women in the household, and the higher levels of access to smartphones and internet among men. The WhatsApp chatbot will be designed to motivate family members, particularly male partners, to take up some of FHWs' routine data and antenatal care work identified through Phase I. I will engage a senior frontline health worker to manage questions that the chatbot is not designed to respond to, taking inspiration from Wizard-of-Oz experiments [49]. Tasks could include completing health surveys, accompanying the expecting mother to antenatal care visits, monitoring nutrition and consumption of iron supplements, and checking weight. I will collect data to determine engagement of family members and the impact on the workload of FHWs. First, I will collect perceived time spent by an FHW on a client when their family members are and are not engaged. Second, I will collect perceived time spent by the family members on antenatal care. Both will be measured using the experience sampling method via a WhatsApp message delivered three times a week. I will also record the WhatsApp chat with the family members to understand their level of engagement over time, the kinds of tasks they took up, and appropriate future engagement strategies. I will supplement the data through interviews with the FHWs, women, and their family members at the beginning, mid-point, and end of the study, to gain a deeper understanding of changing workflows for FHWs and within the home.

Outcomes: The study will contribute an understanding of how automating certain workflows reduces the workload of FHWs and changes how they allocate work time. It will also demonstrate how family members can be engaged in care work through automated chat systems, and the resulting impact on women receiving care.

4.2 Study 2: Data for Making Work Visible

RQ2: How might data collected routinely be leveraged to enable better distribution of work across state and healthcare authorities?

Motivation: My prior work has documented the limited transparency and collaboration across stakeholders around data flows, and the limited use of data to inform decision making at a local or higher level [12]. I will study how the routine data collected by FHWs can be leveraged to make their everyday workflows more visible, and design algorithms for improved distribution of work across government healthcare providers. I will ask the following targeted questions: How can data capture the everyday, invisible work in frontline health, across time and space? How might we visualize this data to enable conversation with diverse stakeholders? How might this data be used by FHWs, local municipal bodies, doctors, supervisors, and policy makers to distribute frontline health work more responsibly and sustainably?

Study Design: Phase I of this study involved a thematic discourse analysis of existing AI systems in frontline health, and interviews with technology practitioners and frontline health workers to understand data flows for ML applications. Building on these findings, Phase II will involve a three-part study with twenty FHWs involving: mobile-based data collection with FHWs to capture their everyday workflows, compilation and visualization of the collected data, and community-based participatory design workshops with

diverse stakeholders including FHWs, community members, policy makers, health authorities, and technologists to determine algorithms for leveraging data flows for the adaptive and fair distribution of work.

I will first conduct a mobile-based data collection exercise with twenty FHWs that mimics their routine data collection. This will involve initial training for a week, after which they will use a mobile application [9] to record their work over one month. All data will be anonymized. Next, I will generate visualizations using this data to represent the work activities that FHWs engaged in, with location and time mapping. I will also aggregate the data to identify the tasks and households that FHWs spent the most time on, and represent these as visualizations. Visualizations will be generated in consultation with FHWs, as well as data scientists. Third, I will share these visualizations in a series of participatory design workshops with local stakeholders—FHWs, community members, the local municipal body, doctors, supervisors, and policymakers. These workshops will be audio and video recorded for data analysis. The goal of these workshops would be to leverage data and visual evidence on the high burden of work to develop algorithms that can help better distribute work across healthcare providers.

Outcomes: Findings will inform the use of data to enable better work distribution across stakeholders. As frontline health moves towards AI-driven workflows, this research will also inform what insights from AI are valuable to stakeholders.

5 RESEARCH CONTRIBUTIONS AND NEXT STEPS

Frontline health work is essential for the operation of a healthcare system that works for women and marginalized populations, yet has remained consistently undervalued. As datafication and automation increasingly enter the healthcare sector, researchers, policymakers, and technologists need to acknowledge and seriously consider the experiences of frontline health workers, and address their frustrations with an underresourced and overburdened system. Grappling with these questions, my dissertation research will offer insight into the design of data-driven and automated systems that reduce the burden on workers, and strengthen frontline health ecologies that are critical to enabling accessible and affordable healthcare for women and communities in resource-constrained settings.

Findings from my completed research have already highlighted some of the challenges and opportunities for technology to augment frontline health work (e.g., [11–13]). I have previously uncovered the limitations of datafication on account of a disconnect between the institutionalized roles and responsibilities, and motivations and on-the-ground workflows of FHWs [11, 12]. This involved extensive fieldwork with FHWs during their visits for data collection in two hundred underserved households in Delhi, India. This led me to distill lessons and insights for improved and *situated* data collection mechanisms across Global Health initiatives, which I intend to draw from in the research proposed above [12]. In continuation of this work, I examined the emerging mobile, internet, and social media practices of FHWs [13]. I studied the online interactions of these workers with their familial units, coworkers, supervisors, (women) patients, and other online communities [13]. I am currently in the

system development and design phase of my dissertation research, which will be followed by implementation and evaluation.

Following graduation, I plan to pursue a career in the nonprofit and policy space, bringing a research and design background to this sector. I am particularly interested in using my background in human-centered design and computing to inform local and state efforts to integrate technology in healthcare workflows. Having benefited from mentorship from HCI researchers, I hope to continue to be part of the HCI research community by submitting practice-oriented research, mentoring students, fostering collaborations between the units where I work and HCI students and faculty, and creating and sharing pathways for social change at a local and global scale.

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